Research Draft: Evaluating the Impact of Nash Learning from Human Feedback (NLHF) on Chain-of-Thought Monitorability and Controllability

Status: Draft / Seeking Collaborators & Feedback

Topic: AI Safety, Social Choice Theory, Post-Training Dynamics, Chain-of-Thought Monitorability

1. Abstract & Motivation

Standard Reinforcement Learning from Human Feedback (RLHF) exhibits fundamental theoretical and practical limitations [Casper23, Halpern26, Maura25]. From a social choice perspective, standard reward modeling structurally acts as a proxy for a **Borda social choice function** [Halpern26], which fails critical voting axioms such as:

- **Condorcet Consistency:** Inability to guarantee the selection of an option that beats every other alternative in pairwise comparisons.
- **Majority Consistency:** Inability to consistently favor options preferred by a strict majority.
- **Independence of Irrelevant Alternatives (IIA):** High vulnerability to ranking shifts caused by uncompetitive or irrelevant options.

These flaws cause RLHF to over-simplify and misrepresent heterogeneous human preferences, resulting in suboptimal alignment and potential representation collapse [Casper23].

Nash Learning from Human Feedback (NLHF) addresses these issues by framing post-training as a two-player zero-sum game [Halpern26, Maura25]. NLHF approximates the **Maximal Lotteries** social choice function, satisfying Condorcet consistency, Majority consistency, and IIA, while preserving preference diversity over mixed outcomes [Halpern26, Maura25].

Given these advantages, NLHF represents a compelling candidate to replace standard RLHF in post-training pipelines. However, adopting a game-theoretic post-training objective raises critical, unexplored questions regarding its downstream impact on model interpretability and safety controls. Inspecting internal model reasoning via Chain-of-Thought (CoT) monitoring has emerged as a primary technique for diagnosing model intent, dishonesty, and potential scheming [Korbak25]. Yet both **CoT monitorability** (the degree to which natural language CoT faithfully reflects internal model intent) and **CoT controllability** (the capability to steer or restrict CoT to safe reasoning pathways) represent promising but highly fragile opportunities for AI safety [Korbak25]. Because CoT monitorability and controllability are acutely sensitive to

changes in optimization dynamics and loss functions [Casper23, Korbak25], replacing RLHF with NLHF may have unintended consequences on whether intermediate CoT reasoning remains transparent, faithful, and steerable.

Key Research Goal: This project investigates how transitioning post-training pipelines from RLHF (Borda-like scalar reward aggregation) to NLHF (Maximal Lotteries / game-theoretic equilibrium) impacts the faithfulness, monitorability, and controllability of Chain-of-Thought reasoning.

2. Primary Research Questions & Hypotheses

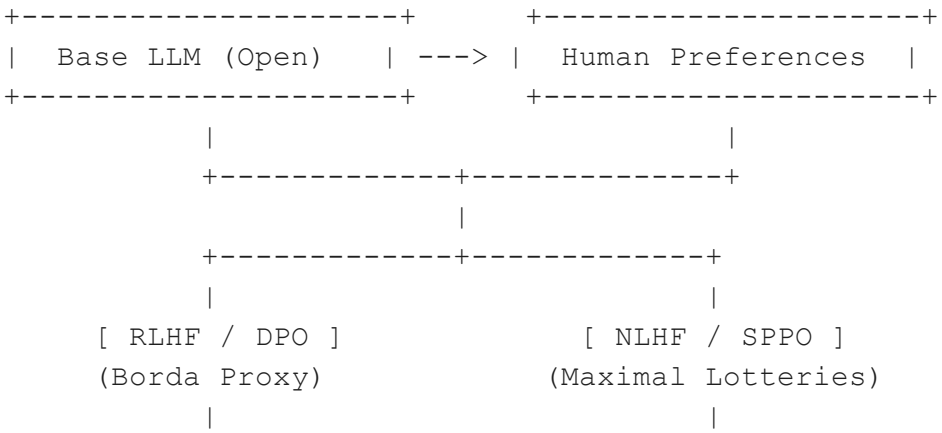
Primary Research Question

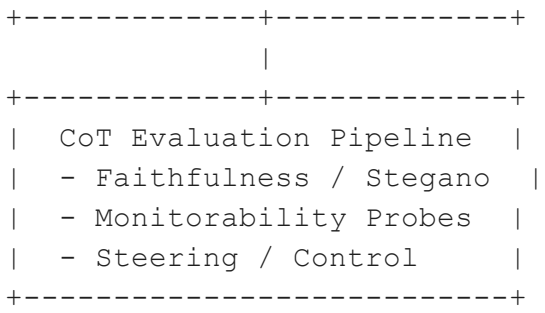
How does shifting post-training alignment from traditional RLHF (Borda-equivalent) to NLHF (Maximal Lotteries-equivalent) affect the faithfulness, monitorability, and controllability of a model's Chain-of-Thought?

Secondary Working Hypotheses

- **H1 (CoT Faithfulness under Preference Heterogeneity):** Because NLHF preserves conflicting preference distributions (as maximal lotteries over candidate responses) rather than forcing them into a single scalar reward function, NLHF-trained models will display lower steganography and obfuscation when responding to ambiguous or multi-preference prompts.
- **H2 (Optimization Pressure & Goodharting):** Game-theoretic equilibrium optimization in NLHF mitigates extreme reward hacking and over-optimization typical of RLHF, preventing late-stage CoT degradation ("reasoning drift" or decoupling of CoT from actual computational trace).
- **H3 (Controllability & Steering Interventions):** Steering interventions (e.g., activation vector addition or system prompt constraints) will remain more effective in NLHF models due to less distorted latent representations.

3. Experimental Plan





Phase 1: Model Training & Post-Training Pipelines

- **Base Models:** Open-weights foundation models (e.g., Llama-3-8B / Qwen-2.5-7B).
- **Datasets:** Multi-preference and standard alignment benchmarks (e.g., UltraFeedback, Anthropic HH-RLHF) representing heterogeneous social preferences.
- **Post-Training Algorithms:**
  - **RLHF Baselines:** Standard PPO / DPO (Borda aggregation proxy).
  - **NLHF Variants:** Self-Play Preference Optimization (SPPO) or Direct Nash Optimization (DNO) (Maximal Lotteries proxy).

Phase 2: CoT Monitorability Evaluation [TBC]

- **Unfaithful Reasoning Detection:** Measure how frequently the model's final answer changes when intermediate CoT steps are perturbed or truncated, following evaluation standards for fragile monitorability.
- **Steganography & Obfuscation Probes:** Evaluate whether models conceal hidden reasoning steps under high optimization pressure.
- **Latent Monitoring & Probing:** Train linear probes on intermediate activations to test whether natural language CoT matches internal latent intent.

Phase 3: CoT Controllability Evaluation [TBC]

- **Constraint Compliance:** Rate adherence to strict step-by-step CoT guidelines under challenging or adversarial contexts.
- **Targeted Steering:** Test susceptibility to activation steering vectors designed to induce or suppress specific reasoning pathways.

4. Expected Contributions

1. **Empirical Benchmarks:** Empirical evaluation of CoT faithfulness, monitorability, and controllability across RLHF and NLHF paradigms.
2. **Social Choice & AI Safety Synthesis:** A bridged evaluation linking social choice properties (Borda vs. Maximal Lotteries) with concrete interpretability metrics.
3. **Open Artifacts:** Codebases, model checkpoints, and benchmark evaluations for public safety research.

5. References

- **[Casper23]** Casper, S., Davies, X., Shi, C., Gilbert, T. K., Scheurer, J., Rando, N., Freedman, R., Korbak, T., & Hadfield-Menell, D. (2023). *Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback*. arXiv preprint arXiv:2307.15217.
- **[Halpern26]** Halpern, D., Micha, E., Procaccia, A. D., Schiffer, B., Shapira, I., & Zhang, S. (2026). *AI Alignment From Social Choice Perspectives*. arXiv preprint arXiv:2606.21550.
- **[Maura25]** Maura-Rivero, R.-R., et al. (2025). *Jackpot! Alignment as a Maximal Lottery*. arXiv preprint arXiv:2501.19266.
- **[Korbak25]** Korbak, T., et al. (2025). *Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety*. arXiv preprint arXiv:2507.11473.